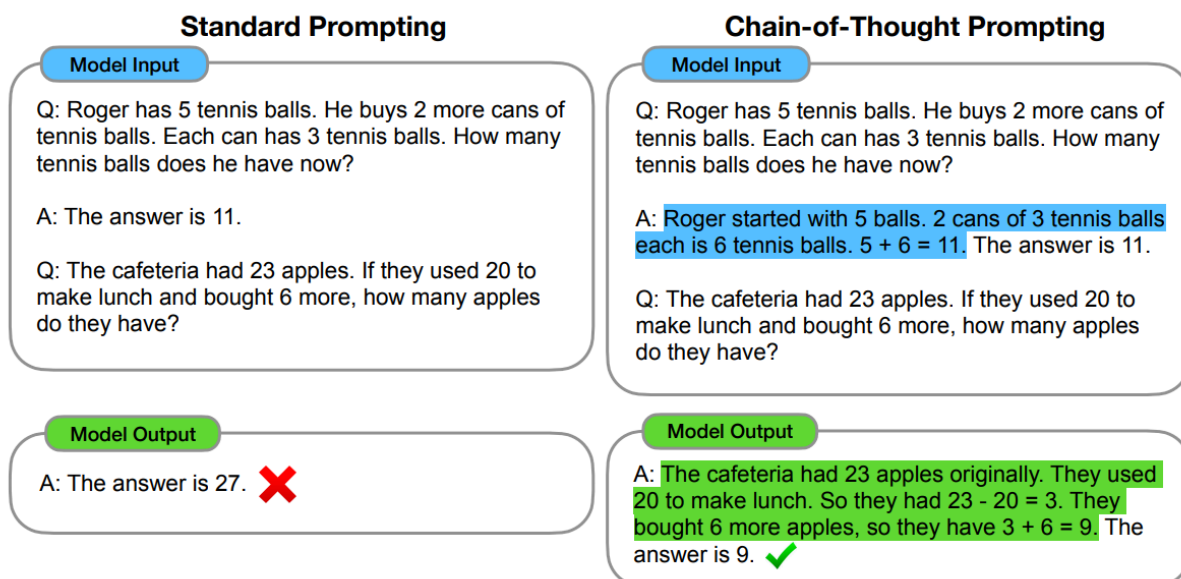


Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

PDF : <https://arxiv.org/pdf/2201.11903>

Abstract



Simply **making intermediate reasoning steps explicit** significantly enhances the complex reasoning capabilities of LLMs

→ *chain of thought prompting*

Introduction

While scaling up language models has shown improved performance and sample efficiency, limitations remain for challenging tasks such as arithmetic, commonsense, and symbolic reasoning.

Ideas & Limitations

1. Arithmetic reasoning can benefit from **generating natural language rationales** that lead to the final answer
 - Creating a large set of high quality rationales is costly
2. In-context **few-shot learning via prompting** is helpful for simple tasks
 - It yields poor results on tasks requiring reasoning & shows limited improvement with scaling up models.

Approach

Provide a **prompt with triples** : <input, ***chain of thought***, output>

Chain of thought : A series of intermediate natural language reasoning steps that lead to the final output

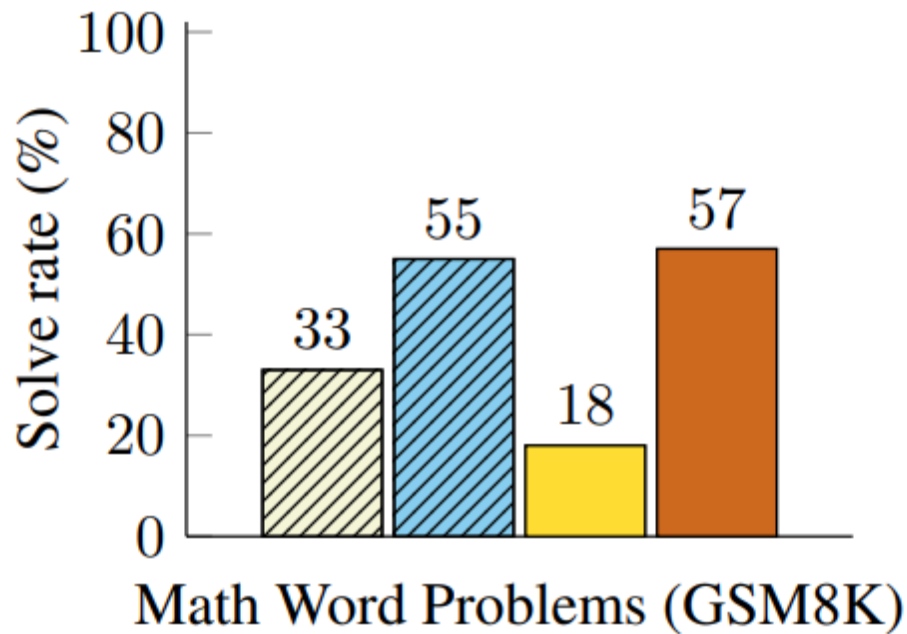
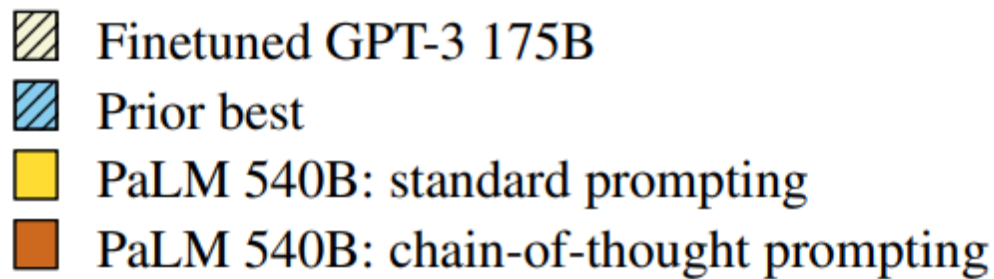


Figure 2: PaLM 540B uses chain-of-thought prompting to achieve new state-of-the-art performance on the GSM8K benchmark of math word problems. Finetuned GPT-3 and prior best are from Cobbe et al. (2021).

→ This work underscores how large language models can learn via a few examples with natural language data about the task

Chain-of-Thought Prompting

Core Concept

It is natural to decompose complicated reasoning tasks into each simple intermediate steps

→ **chain-of thought mimics** this process

Properties of chain-of-thought

1. **Decomposition** : Allows models to decompose multi-step problems into intermediate steps
2. **Interpretability** : Provides an interpretable window into the behavior of the model
→ Helps identify where the reasoning path went wrong
3. **Applicability** : Potentially applicable to any tasks that humans can solve via language
4. **Ease of Use** : Readily elicited in large language models simply by including examples int the prompt

Arithmetic Reasoning

LLM often struggles from arithmetic reasoning

Experimental Setup

Benchmarks

GSM8K, SVAMP, ASDiv, AQuA, MAWPS

Standard prompting

Few-shot prompting, given in-context exemplars of input-output pairs (questions, answers), before outputting a prediction

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. 

Chain-of-thought prompting

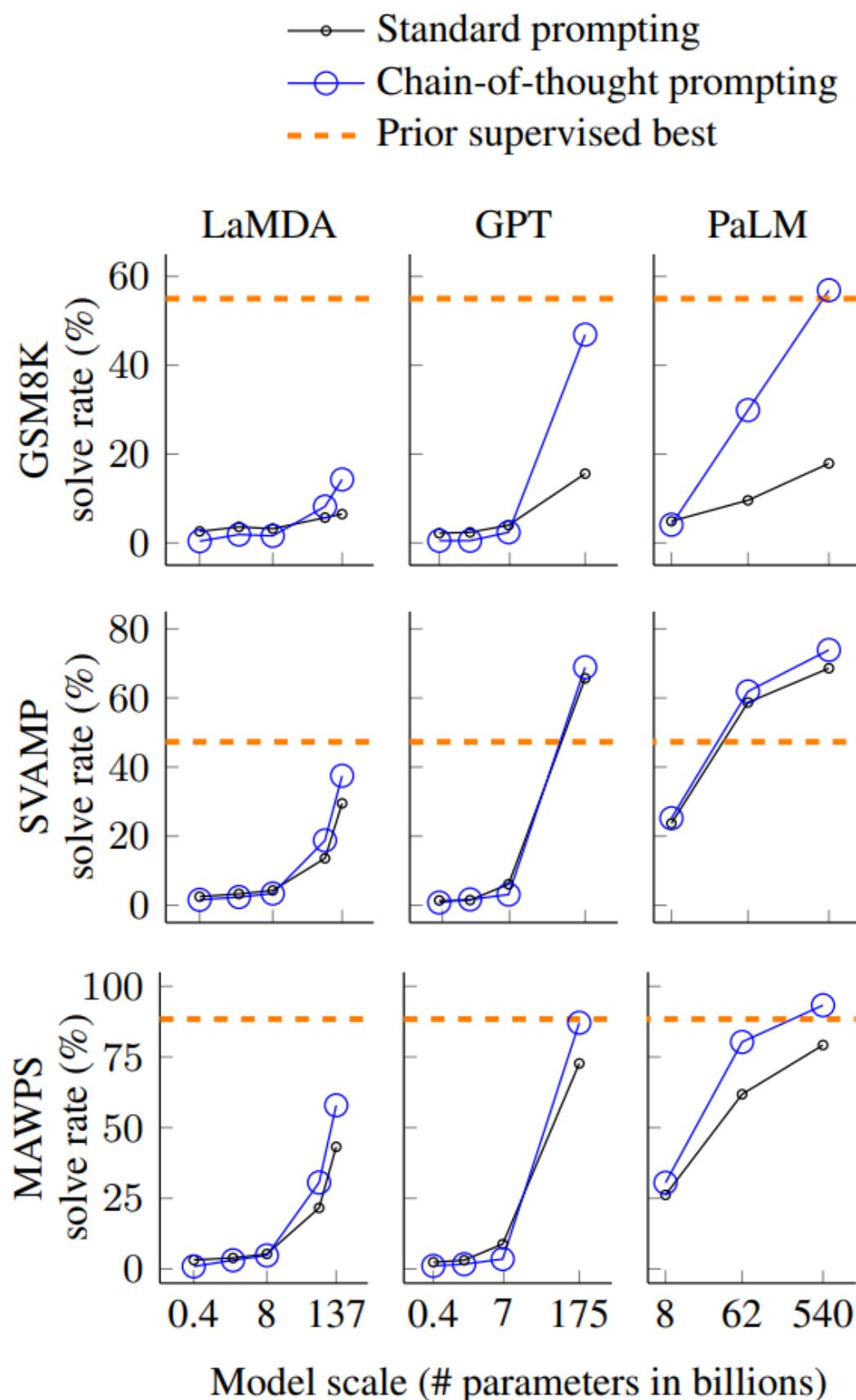
Augment each exemplar in few-shot prompting with a chain of thought for an associated answer

Math Word Problems (free response) Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now? A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.	Math Word Problems (multiple choice) Q: How many keystrokes are needed to type the numbers from 1 to 500? Answer Choices: (a) 1156 (b) 1392 (c) 1480 (d) 1562 (e) 1788 A: There are 9 one-digit numbers from 1 to 9. There are 90 two-digit numbers from 10 to 99. There are 401 three-digit numbers from 100 to 500. $9 + 90(2) + 401(3) = 1392$. The answer is (b).	CSQA (commonsense) Q: Sammy wanted to go to where the people were. Where might he go? Options: (a) race track (b) populated areas (c) desert (d) apartment (e) roadblock A: The answer must be a place with a lot of people. Race tracks, desert, apartments, and roadblocks don't have a lot of people, but populated areas do. So the answer is (b).
StrategyQA Q: Yes or no: Would a pear sink in water? A: The density of a pear is about 0.6 g/cm^3, which is less than water. Thus, a pear would float. So the answer is no.	Date Understanding Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY? A: One day after 06/01/1943 is 06/02/1943, so today is 06/02/1943. 10 days before today is 05/23/1943. So the answer is 05/23/1943.	Sports Understanding Q: Is the following sentence plausible? "Joao Moutinho caught the screen pass in the NFC championship." A: Joao Moutinho is a soccer player. The NFC championship is part of American football, not soccer. So the answer is no.
SayCan (Instructing a robot) Human: How would you bring me something that isn't a fruit? Explanation: the user wants something to eat that isn't a fruit. An energy bar is not a fruit, so I will bring the user an energy bar. Plan: 1. find(energy bar) 2. pick(energy bar) 3. find(user) 4. put(energy bar) 5. done().	Last Letter Concatenation Q: Take the last letters of the words in "Lady Gaga" and concatenate them. A: The last letter of "Lady" is "y". The last letter of "Gaga" is "a". Concatenating them is "ya". So the answer is ya.	Coin Flip (state tracking) Q: A coin is heads up. Maybelle flips the coin. Shalonda does not flip the coin. Is the coin still heads up? A: The coin was flipped by Maybelle. So the coin was flipped 1 time, which is an odd number. The coin started heads up, so after an odd number of flips, it will be tails up. So the answer is no.

Language models

GPT-3, LaMDA, PaLM, UL2 20B, Codex

Results



Takeaways

1. Emergent ability of model scale:

→ Smaller models produce fluent but illogical chains of thought

2. Larger gains for complex problems:

→ Performance gains more significant for complicated reasoning tasks

3. Comparable to SOTA:

→ Achieves performance comparable to SOTA models via prompting alone

Why Chain-of-Thought Works (Error Analysis)

- Examination of 50 **correct answers**:
 - 48/50 → Correct logic / Correct answer
 - 2/50 → Incorrect logic / **Coincidentally** correct answer
- Examination of 50 **incorrect answers**:
 - 23/50 → Correct logic with minor mistakes
 - 27/50 → Incorrect logic with major errors

Scaling & Chain-of-Thought

- Scaling fixes a large portion of **one-step missing and semantic understanding errors**

Ablation Study

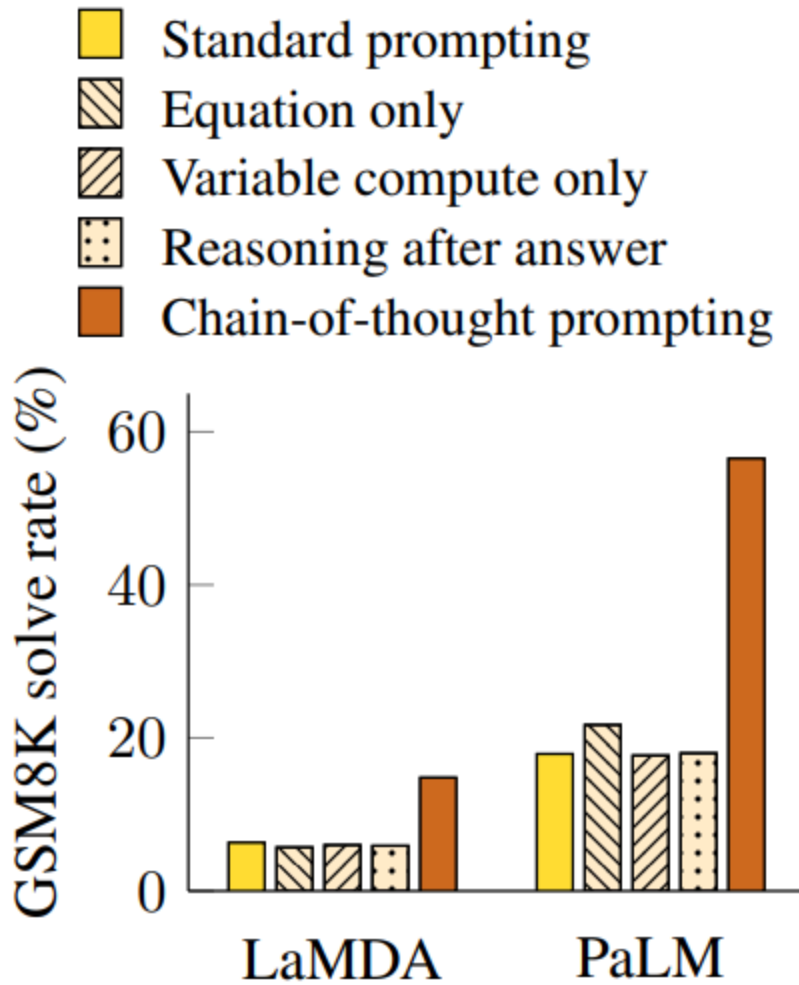


Figure 5: Ablation study for different variations of prompting using LaMDA 137B and PaLM 540B. Results for other datasets are given in Appendix Table 6 and Table 7.

Equation only

- **Hypothesis:** Chain-of-thought might **produce the mathematical equation to be evaluated**
- **Method:** Prompted to **output only a mathematical equation** before the answer

- **Result:**
 - **Little to no enhancement on complex tasks:** Direct translation of semantics into equations is **too challenging** without reasoning steps
 - **Effective for simpler tasks:** For one- or two-step problems, equation-only prompts do improve performance

Variable compute only

- **Hypothesis:** Chain-of-thought might allow the model to **spend more computation on harder problems**
- **Method:** Prompted to **output only a sequence of dots** equal to the number of characters in the equation
- **Result:**
 - **Variable computation alone is insufficient:** Success is not solely due to increased computation tokens

Chain of thought after answer

- **Hypothesis:** Chain of thought might simply **help access knowledge acquired during pretraining**
- **Method:** Chain of thought is given **only after the final answer**
- **Result:**
 - **Performance remains at baseline:** This suggest that the **sequential reasoning process itself is critical**, beyond just activating knowledge

Robustness of Chain of Thought

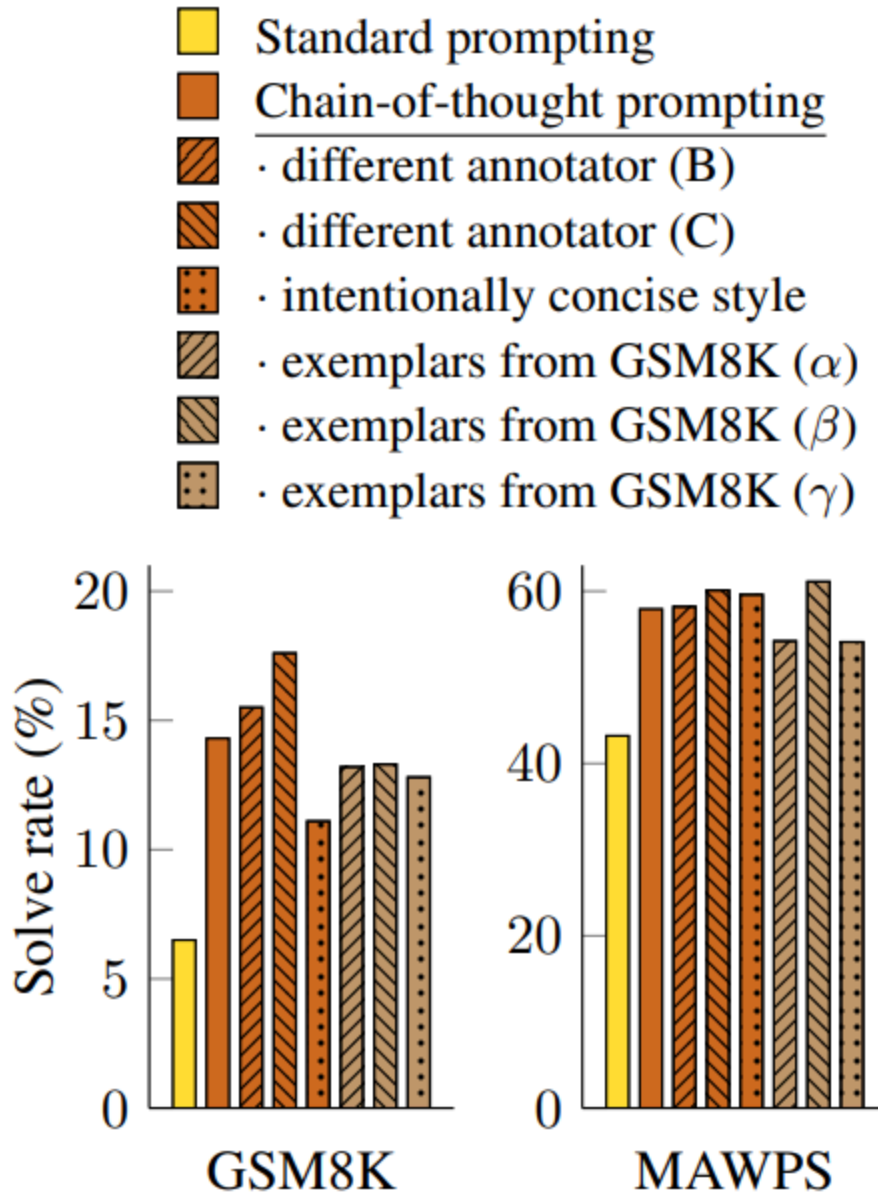


Figure 6: Chain-of-thought prompting has variance for different prompt examples (as expected) but outperforms standard prompting for various annotators as well as for different exemplars.

Sensitivity to Exemplars

1. Linguistic style (Annotators)

- Despite some variance among different annotations, all sets of prompts outperform the standard baseline by a large margin
 - **Robust to the linguistic style:** Success does not depend on a particular style

2. Specific exemplars

- Successful chain-of-thought prompting works equally well with other sets of exemplars
 - **Robust to specific exemplar choices**

Commonsense Reasoning

Benchmarks

CSQA, StrategyQA, Date, Sports, SayCan

Prompts

Follow the same experimental setup as the prior section

Results

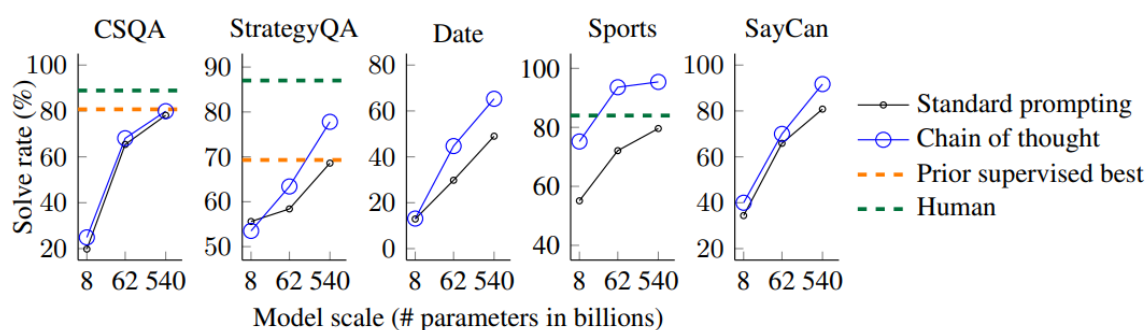


Figure 7: Chain-of-thought prompting also improves the commonsense reasoning abilities of language models. The language model shown here is PaLM. Prior best numbers are from the leaderboards of CSQA (Talmor et al., 2019) and StrategyQA (Geva et al., 2021) (single-model only, as of May 5, 2022). Additional results using various sizes of LaMDA, GPT-3, and PaLM are shown in Table 4.

- Scaling up model size improves the performance of standard prompting
- Chain-of-thought prompting yields further gains
 - **Conclusion:** Chain-of-thought prompting **improves performance on tasks requiring a range of commonsense reasoning abilities**

Symbolic Reasoning

Tasks

- **Last letter concatenation**
 - Concatenate the last letters of words in a name
 - e.g. *"Amy Brown"* → *"yn"*
- **Coin flip**
 - Ask the model to answer whether a coin is still heads up after people either flip or don't flip the coin
 - e.g. *"A coin is heads up. Phoebe flips the coin. Osvaldo does not flip the coin. Is the coin still heads up?"* → *"no"*

In-domain test : examples have **the same number** of steps as training/few-shot exemplars

Out-of-domain test : examples have **more steps** than those in the exemplars

Generalization Test: For both tasks, the model only sees exemplars with **two** items (words/flips)

→ **Evaluated on** sequences of **three or four** items.

Results

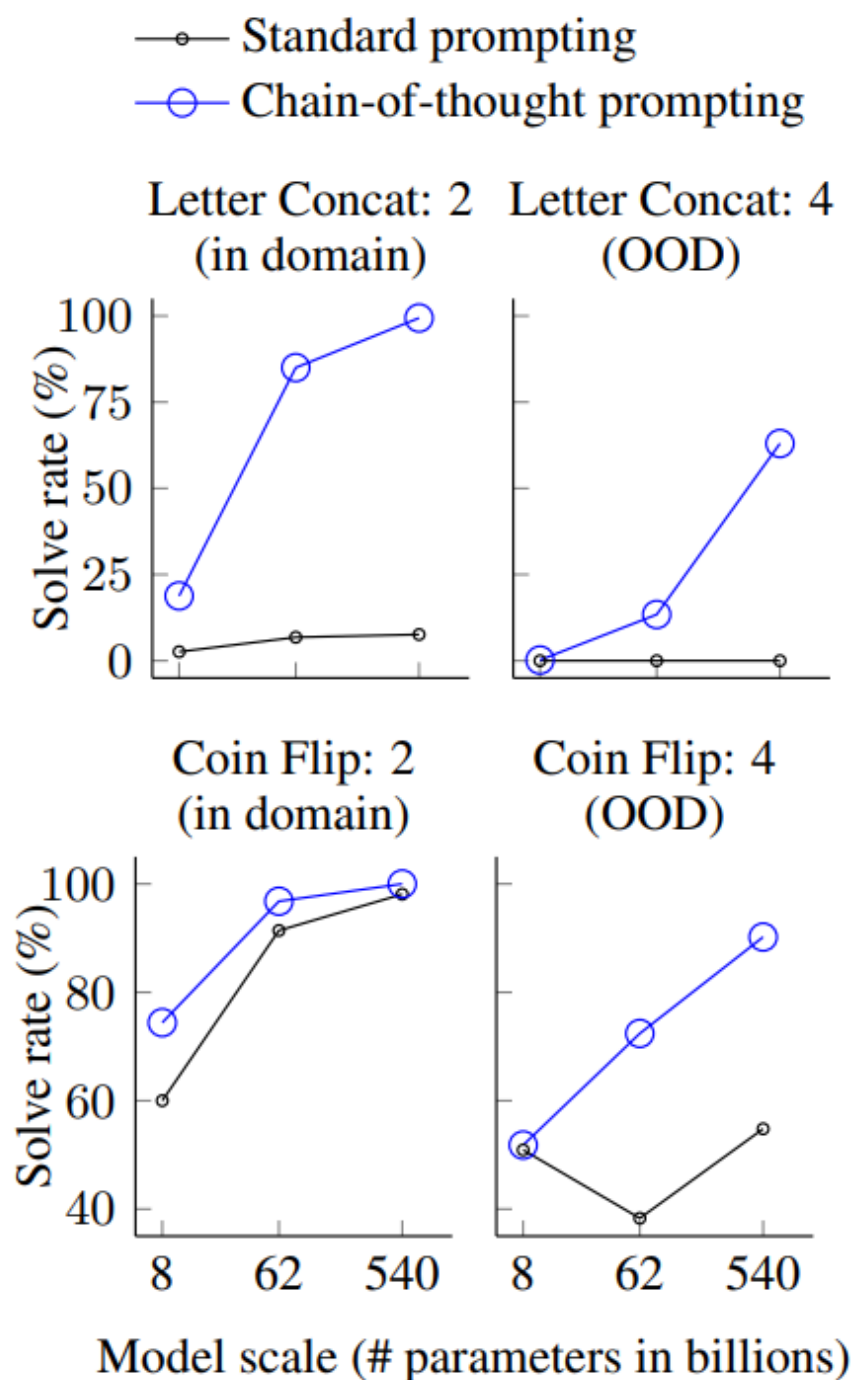


Figure 8: Using chain-of-thought prompting facilitates generalization to longer sequences in two symbolic reasoning tasks.

In-domain Evaluations

- **PaLM 540B achieves** almost 100% solve rates.
 - **Note:** Small models still fail to perform abstract manipulations **despite the simplicity of the tasks.**

Out-of-domain (OOD) Evaluations

- **Standard prompting fails completely.**
 - **Conclusion:** Chain-of-thought prompting facilitates **length generalization** beyond seen examples for sufficiently large models.

Discussion

Summary & Future Questions

- **Performance:** Chain-of-thought prompting **improves performance by a large margin** on arithmetic, commonsense, and symbolic reasoning **via simple prompt engineering** (without finetuning)
- **Potential:** Standard prompting only provides a **lower bound** on the capabilities of LLMs
 - Q1: How much more can **reasoning ability improve with further model scaling?**
 - Q2: **What other prompting methods** might expand the range of solvable tasks?

Limitations

- **True Reasoning :** Is the neural network actually “reasoning,” or just mimicking the prompt?
- **Cost:** Annotation costs can be prohibitive for finetuning (unlike few-shot prompting)
- **Reliability:** No guarantee of correct reasoning paths
- **Serving Cost:** Chain-of-thought reasoning emerges only in large models, making it costly to serve compared to smaller models

Related Work

Using intermediate steps to solve reasoning problems

- using natural language rationales to solve math word problems through **a series of intermediate steps** (Ling et al. (2017))
- extended by **creating a larger dataset and using it to finetune** a pretrained model (Cobbe et al. (2021))
- via **line-to-line prediction**, predicted the final outputs of Python programs (Nye et al. (2021))

Few-shot prompting

- **few-shot prompting** (Brown et al. (2020))
- automatically **learning prompts** (Lester et al. (2021))
- giving models **instructions describing a task** (Wei et al. (2022))

Chain-of-thought focuses on **augmenting the output of language models**

Conclusion

- **Chain-of-thought prompting:** A simple and broadly applicable method for enhancing reasoning in language models
- **Key Finding:** Chain-of-thought prompting is an emergent property of **model scale** (specifically in large language models)
- **Future Outlook:** Broadening the range of reasoning tasks will inspire further work on language-based approaches to reasoning